

Problem Solving in Engineering (IP-3)

Module 3 Workbook: Serenity VR

Causal Loop Diagrams & Leverage

Identifying Feedback Loops

Reinforcing Loop (R) - "The Vicious Cycle of Avoidance and Worsening Symptoms"

Path: Patient experiences anxiety → Long wait times/High costs of therapy → Patient avoids seeking help → Anxiety goes untreated → Symptoms worsen over time → Therapy becomes even more expensive and intimidating → Avoidance increases.

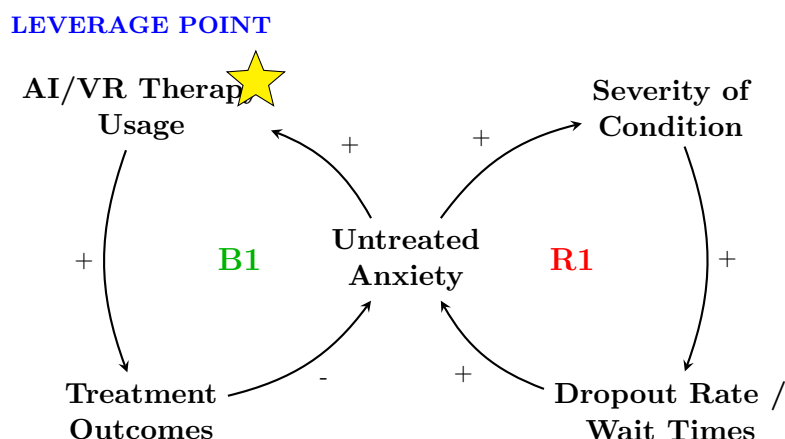
Why it's reinforcing: The lack of accessible therapy causes the condition to deteriorate, which ironically makes the patient even less likely or able to seek the now-more-intense therapy they need. The problem pushes itself in the same worsening direction.

Balancing Loop (B) - "The Therapy Correction Loop"

Path: Patient experiences anxiety → Patient accesses professional therapy → Receives clinical guidance → Applies techniques → Symptoms reduce → Need for therapy decreases.

Current state: This balancing loop is currently **broken or too weak** in India because the link between "experiences anxiety" and "accesses professional therapy" is severed by the 0.3/100,000 psychiatrist shortage.

Causal Loop Diagram



Leverage Point

Location in the loop: The connection between "Patient experiences anxiety" and "Access to therapeutic guidance."

The Intervention (How Might We): By introducing **Serenity VR** (AI/VR Biofeedback Therapy), we inject a strong, highly accessible pathway directly into the Balancing Loop.

Instead of waiting 2–8 weeks to see a professional, the patient puts on a VR headset at home and immediately receives AI-driven, biofeedback-adapted therapy.

Why it shifts the system: This small change (giving the patient immediate access to automated, adaptive therapy) completely bypasses the human supply constraint. It stops the Reinforcing Loop (worsening symptoms from waiting) and strengthens the Balancing Loop (rapid symptom reduction), stabilizing the system.

Matching the System Archetype

1. Which archetype best describes your system problem?

Limits to Growth (and elements of *Shifting the Burden*).

2. Why does this archetype fit your problem?

The "Engine of Growth" (Reinforcing): Post-pandemic, there has been a massive increase in mental health awareness and a drive to seek therapy. More people are trying to get help, which is fundamentally a positive growth trend.

The "Limit to Success" (Balancing constraint): The physical capacity of the health-care system. India has a hard limit of ~0.3 psychiatrists per 100,000 people.

How the archetype plays out: As mental health awareness grows and more people seek help, the system rapidly hits its physical limit (the capacity of existing professionals). As a result:

- Wait times skyrocket (2–8 weeks).
- Quality of care drops (sessions become rushed, 15–30 patients a day).
- Costs increase due to demand.
- Patients get frustrated, outcomes suffer, and they drop out.

The Fix: Simply trying to "push" the engine of growth (telling more people to go to therapy) or "working harder" (doctors seeing 40 patients a day) will fail, just like the dieting bind or People Express airlines. The only way to solve a *Limits to Growth* archetype is to **remove or elevate the limit**.

Serenity VR elevates the limit by using AI and VR to deliver adaptive therapy, effectively multiplying the capacity of the mental health system without needing to magically train 40,000 new psychiatrists overnight.